

EDUCATION

- 2026 **Doctor of Philosophy in Environmental Engineering** Berkeley, CA
 University of California, Berkeley
 Dissertation: Engineering Environmental Policy for Advancing Health, Climate, and Equity Goals
- 2018 **Master of Engineering in Civil and Environmental Engineering** Cambridge, MA
 Massachusetts Institute of Technology
 Thesis: The Phase Separation Inlet for Droplets, Ice Residuals, and Interstitial Aerosols
- 2017 **Bachelor of Science in Earth, Atmospheric, and Planetary Sciences** Cambridge, MA
 Massachusetts Institute of Technology
 Thesis: Characterization of a 3D Printed Pumped Counterflow Virtual Impactor and an Aerodynamic Lens Concentrator
 Minor: Atmospheric Chemistry; Concentration: Music and Theater Arts

PROFESSIONAL APPOINTMENTS

- 2026 - **Postdoctoral Scholar**, Northwestern University Evanston, IL
present (Mentor: Prof. Daniel Horton)
- 2021-26 **Graduate Student Researcher & Fellow**, University of California, Berkeley Berkeley, CA
 (Advisors: Profs. Joshua Apte and Rachel Morello-Frosch)
- 2018-21 **Air Quality Consultant & Senior Air Quality Consultant**, Ramboll San Francisco, CA
 Area of expertise: emissions inventory development and health risk analysis.
- 2015-18 **Undergraduate & Graduate Research Fellow**, MIT Cambridge, MA
 (Advisor: Prof. Daniel Cziczo).
- 2014 **Undergraduate Research Fellow**, MIT Cambridge, MA
 (Advisor: Prof. Noelle Selin)

PUBLICATIONS

- In prep.* **Koolik, L. H.**, Swayne, M., Sadd, J., Pastor, M., and Morello-Frosch, R.: Challenges and Recommendations for Quantifying Progress in California's Community Air Protection Programs. *In prep.*
- In press* Manchanda, C. & **Koolik, L. H.** (co-first), Ünal, A., Fung, I., Marshall, J. D., Morello-Frosch, R., Turner, A. J., Harley, R. A., and Apte, J.S.: Inverse Modeling Identifies Efficient Emission Control Strategy for Mitigating PM_{2.5} Pollution. In press at *Env. Sci. Technol.*
- 2026 Bekbulat, B., Sharma, S., Apte, J. S., Bullard, R. D., **Koolik, L. H.**, Min, E., Morello-Frosch, R., Pastor, M., Patterson, R., Robinson, A. L., Salgado, M., Unal, A., Wedekind, N., Marshall, J. D.: PM_{2.5} Air Pollution Inequities in the U.S. by Sector and State: Past Trajectories and Future Directions, *Sci. Adv.*, **12** (24), eaed7425, 2026. DOI: 10.1126/sciadv.aed7425. Available at: <https://doi.org/10.1126/sciadv.aed7425>.

- 2026 **Koolik, L. H.**, Speizer, S., Rong, C., Chambliss, S., Marshall, J. D., Morello-Frosch, R., Tessum, C., and Apte, J. S.: Methodological Design Choices Can Affect Air Pollution Exposure Disparity Estimates: A Case Study on California’s Agricultural Sector, *Env. Sci. Technol.*, **60** (6), 4753–4763, 2026. DOI: 10.1021/acs.est.5c10796. Available at: <https://doi.org/10.1021/acs.est.5c10796>.
- 2026 Marshall, J. D., **Koolik, L. H.**, Ünal, A., Morello-Frosch, R., and Apte, J. S.: Advancing Methods and Models that Promote Equity in Air Quality, *Annu. Rev. Publ. Health*, **47**, 2026. Available at: <https://doi.org/10.1146/annurev-publhealth-091824-125106>.
- 2025 **Koolik, L. H.**, Bullard, R. D., Min, E., Morello-Frosch, R., Salgado, M., Patterson, R., Wedekind, N., Marshall, J. D., and Apte, J. S.: Eliminating Air Pollution Disparities Requires More than Emission Reduction. *Proc. Natl. Acad. Sci. USA*, **122** (50), e2505888122, 2025. Available at: <https://doi.org/10.1073/pnas.2505888122>.
Press Coverage: [1]
- 2024 **Koolik, L. H.**, Alvarado, Á., Budahn, A., Plummer, L., Marshall, J. D., and Apte, J. S.: PM_{2.5} Exposure Disparities Persist Despite Strict Vehicle Emissions Controls in California, *Sci. Adv.*, **10**, eadn8544, 2024. Available at: <https://doi.org/10.1126/sciadv.adn8544>.
Press Coverage: [1], [2], [3], [4], [5]
- 2022 **Koolik, L.**, Roesch, M., Dameto de Espana, C., Rapp, C. N., Franco Deloya, L. J., Shen, C., Hallar, A. G., McCubbin, I. B., and Cziczo, D. J.: A Phase Separation Inlet for Droplets, Ice Residuals, and Interstitial Aerosol Particles, *Atmos. Meas. Tech.*, **15**, 3213–3222, 2022. Available at: <https://doi.org/10.5194/amt-15-3213-2022>.

AWARDS AND FELLOWSHIPS

- 2025
- Philomathia Graduate Student Fellowship in the Environmental Sciences ([link](#))
 - Joan Daisey Air Quality Research Award
 - Lau Graduate Fellowship in Climate Equity
 - Selected for NextProf Nexus 2025
- 2024
- American Geophysical Union Outstanding Student Presentation Award ([link](#))
- 2023
- Hearts to Humanity Eternal (H2H8) Graduate Research Fellowship ([link](#))
 - Health Effects Institute Jane Warren Award ([link](#))
- 2021
- University of California, Berkeley Chancellor’s Fellowship for Graduate Study
- 2019
- Ramboll Extraordinary Individual Contribution to the Business Unit Award
- 2016
- MIT Department of Earth, Atmospheric, and Planetary Science Achievement Award

MENTORSHIP AND ADVISING

Graduate Student Research Collaboration

- Simone Speizer (2024–present): “Methodological Design Choices Can Affect Air Pollution Exposure Disparity Estimates: A Case Study on California’s Agricultural Sector.” *Manuscript published in Environmental Science & Technology*.
- Cassidy Barrientos (2024–present): “Historical Trends in Exposure Equity Associated with California’s Cap and Trade Program.” *Manuscript in preparation*.
- Lucas Rojas Mendoza (2023–present): “Analyzing Sources and Scales of Air Pollution Disparities in the US and California: Examining Urban-Rural Emission Interactions in disadvantaged communities.” Poster presented at the

American Geophysical Union Fall Meeting 2023, San Francisco, CA. December 11-15, 2023. *Manuscript in preparation.*

Undergraduate Student Research Mentorship

- Meghana Raj (2024–2026): “Comparing Methodologies for Air Pollution Health Impact Assessments in Open-Source Modeling for Equity in California.” Direct contributions to the modeling pipeline currently in use by California state agencies.
- Benjamin Salop (2024–2025): “Future-Proofing Open-Source, Accessible Air Pollution Modeling Pipelines for Increased Usability.” Direct contributions to the modeling pipeline currently in use by California state agencies.
- Amy Yao (2024): “Developing Automated Techniques for Processing Complex Population Data.” Direct contributions to the modeling pipeline currently in use by California state agencies.
- Clara Rong (2023–2024): “Decomposing California’s Agricultural Sector for Insights Towards Equitable Air Quality.” Poster presented at the American Geophysical Union Fall Meeting 2023, San Francisco, CA. December 11–15, 2023. *Manuscript published in Environmental Science & Technology.*
- Thomas Le (2022–2023): “Increasing Accessibility for Modeling Point Source Emissions.” Emissions processing pipeline developed is currently in use by state agencies.

Other Mentorship

- MIT Terrascope Alumni Mentor (2022–present): provide technical support for undergraduate program that challenges freshmen to develop socio-politically informed engineering solutions to global environmental problems.
- Berkeley Graduate Women in Engineering & Society of Women Engineering Mentor (2022–2026): provide research and career advice to undergraduate women and non-binary engineers.

Advisory Roles

- UCLA Environmental Science Senior Practicum (2024 & 2025): provide introductory training resources and ongoing support for air pollution modeling efforts by undergraduate student research teams advised by Prof. Pablo Saide.
- Community Health and Environmental Impacts Section of the California Office of Environmental Health Hazard Assessment (2022–present): provide ongoing technical support and code development for open-access model developed.

INVITED PRESENTATIONS

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| 2025 | <ul style="list-style-type: none">• “Triple Wins, not Trade-offs: Investigating environmental policies at the climate-health-equity nexus.” Research seminar at Oberlin College for Environmental Studies and Science program, Oberlin, OH. December 2, 2025. |
| 2024 | <ul style="list-style-type: none">• Featured presenter and panelist at American Geophysical Union GeoHealth Outstanding Student Presentation Award-Winning Research: Exploring Equity and Emission Impacts in GeoHealth. Virtual. July 19, 2024.• “For Exposure to PM_{2.5} from California’s On-Road Mobile Sources, Relative Disparities by Race-Ethnicity Remain Even After Decades of Emissions Controls.” Oral presentation at the 2024 Joint American Geophysical Union/American Meteorological Society Showcase. Virtual. April 3, 2024. |
| 2023 | <ul style="list-style-type: none">• EJ-AIR Workshop: Using Air Pollution Data and Models for Environmental Justice, Berkeley, CA. December 7–9, 2023. |

- “Racial-Ethnic Disparities in Exposure to PM_{2.5} from California’s On-Road Mobile Sources Remain After Decades of Emissions Controls.” Featured talk in the Jane Warren Award Plenary at the Health Effects Institute Annual Conference, Boston, MA. April 28–May 3, 2023.
- PAVITRA Project Launch and Capacity Building Workshop, Bengaluru, India. March 2–6, 2023.

CONFERENCE PRESENTATIONS

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| 2026 | <ul style="list-style-type: none"> • “A Bayesian Inverse Modeling Approach to Achieving Triple Wins in Air Quality, Climate, and Equity”. Oral presentation at the European Geophysical Union General Assembly, Vienna, Austria. May 3-8, 2026. |
| 2025 | <ul style="list-style-type: none"> • “Inverting Environmental Policy: A Bayesian Framework for Achieving Triple Wins in Air Quality, Climate, and Equity.” Poster presentation at the Health Effects Institute Annual Conference, Austin, TX. May 4–6, 2025. |
| 2024 | <ul style="list-style-type: none"> • “A Conceptual Framework Towards Equity-Oriented Decision-Making in Air Pollution.” Poster presentation at American Geophysical Union Fall Meeting, Washington D.C. December 9–13, 2024. • “For Exposure to PM_{2.5} from California’s On-Road Mobile Sources, Relative Disparities by Race-Ethnicity Remain Even After Decades of Emissions Controls.” Oral presentation at International Society for Environmental Epidemiology Annual Conference, Santiago, Chile. August 25–28, 2024. • “A Conceptual Framework Towards Equity-Oriented Decision-Making in Air Pollution.” Poster presentation at Health Effects Institute Annual Conference, Philadelphia, PA. April 28–30, 2024. |
| 2023 | <ul style="list-style-type: none"> • “For Exposure to PM_{2.5} from California’s On-Road Mobile Sources, Relative Disparities by Race-Ethnicity Remain Even After Decades of Emissions Controls.” Oral presentation at American Geophysical Union Fall Meeting, San Francisco, CA. December 11–15, 2024. Winner of the 2023 Outstanding Student Presentation Award (link). • “Racial-Ethnic Disparities in Exposure to PM_{2.5} from California’s On-Road Mobile Sources Remain After Decades of Emissions Controls.” Poster presentation at the Health Effects Institute Annual Conference, Boston, MA. April 28–May 3, 2023. Winner of the 2023 Jane Warren Award (link). |

TEACHING EXPERIENCE

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| 2026 | <ul style="list-style-type: none"> • Discipline Cluster Leader for UC Berkeley Spring Teaching Conference for First-Time Student Instructors. |
| 2025 | <ul style="list-style-type: none"> • Discipline Cluster Leader for UC Berkeley Fall Teaching Conference for First-Time Student Instructors. |
| 2022 | <ul style="list-style-type: none"> • Graduate Student Instructor for Berkeley School of Public Health graduate-level course on Exposure Assessments and Controls (PBHLTH 270A). |
| 2017,
2018 | <ul style="list-style-type: none"> • Teaching Assistant for MIT’s Discover Earth, Atmospheric, and Planetary Sciences Extreme Weather freshman program. |
| 2015 | <ul style="list-style-type: none"> • 7th grade Chemistry Instructor for MIT’s Office of Engineering Outreach Program’s middle school STEM program. • Volunteer and guest teacher as part of MIT and Teach for America’s Four Weeks for America teaching program. |

- 2014
- Undergraduate Teaching Fellow for the Solving Complex Problems course through MIT’s “Mission 2018” cohort of the Terrascope program.

SERVICE AND OUTREACH

Academic and Professional Service

- American Geophysical Union GeoHealth Early Career Committee.
- Convener for session on “Data-Driven Policy for Air Quality and Climate Resilience” at the American Geophysical Union Fall 2026 meeting.
- Early career convener for session on “Innovative Models and Tools to Precisely Quantify and Identify Solutions for Air Pollution Exposure Inequality” at the American Geophysical Union Fall 2024 meeting.
- Peer review for *Environmental Science & Technology* and *GeoHealth*.
- 2023 Environmental Engineering Seminar Organization Committee.

Community Service and Outreach

- Lead coordinator for a series of air filtration workshops for La Clinica de la Raza, Community Resources for Science, and Stockton Unified School District (2023–2026).
- Panelist for the NSF-funded CARE panel to share my career pathway to underrepresented students in the Chicago area (April 7, 2022).

Open-Source Software and Resource Development

- Estimating Concentrations and Health Outcomes: Automated ISRM Resource (ECHO-AIR): Lead engineer of a fully open-source model designed to increase accessibility in high-resolution air pollution modeling. See more at: <https://echo-air-model.github.io/>
- Collection of research graphics and slide templates available on my personal website: <https://lkoolik.github.io/>